

Week 9

WACC

This week, we are going to look at the followings:

Debt vs Preference Shares vs Ordinary Shares - SLO3, 5

Weighted Average Cost of Capital (WACC) - SLO3, 5

# Learning Objectives

At the end of this module you will:

SLO3: demonstrate an understanding of financial risk, its measurement, and its relationship to return

SLO5: assess a firm's financing options and propose a strategy for raising capital

# Cost of Debt

- Interest rate required on new debt issuance (i.e., yield to maturity on outstanding debt)
- Adjust for the tax deductibility of interest expense

*Aftertax cost of debt = (1 - tax rate)  $\times$  borrowing rate*

$$= R_B \times (1 - T_C)$$

## **Example: cost of debt**

- Shanken Corp. issued a 30-year, 7 percent semiannual bond 7 years ago. The bond currently sells for 108 percent of its face value. The company's tax rate is 35 percent.
- **(1) What is the pretax cost of debt?**
- **(2) What is the aftertax cost of debt?**

## *Example: cost of debt*

(1) The pretax cost of debt is the YTM of the company's bonds,  
so:

$$P_0 = \$1,080 = \$35(PVIFA_{R\%,46}) + \$1,000(PVIF_{R\%,46})$$

$$R = 3.167\%$$

$$YTM = 2 \times 3.167\% = 6.33\%$$

(2) The aftertax cost of debt is:

$$R_B = .0633(1 - .35) = .0412 \text{ or } 4.12\%$$

# Cost of Preferred Stock

- Preferred stock is a perpetuity, so its price is equal to the coupon paid divided by the current required return.
- Rearranging, the cost of preferred stock is:
  - $R_p = C / PV$

## *Example :cost of preferred stock*

- Suppose a share of the preferred stock of Polytech, Inc. is selling at \$17.16 and pays a dividend of \$1.50 per year.
- What is the cost of preferred stock?
  - ▶  $R_p = C / PV$

$$8.7\% = 1.50 / 17.16$$

# The Weighted Average Cost of Capital

► The Weighted Average Cost of Capital is given by:

$$R_{WACC} = \frac{\text{Equity}}{\text{Equity} + \text{Debt}} \times R_{\text{Equity}} + \frac{\text{Debt}}{\text{Equity} + \text{Debt}} \times R_{\text{Debt}} \times (1 - T_C)$$

$$R_{WACC} = \frac{S}{S + B} \times R_S + \frac{B}{S + B} \times R_B \times (1 - T_C)$$

- Because interest expense is tax-deductible, we multiply the last term by  $(1 - T_C)$ .

# The Weighted Average Cost of Capital

- ▶ First, we estimate the cost of equity and the cost of debt.
  - We estimate an equity beta to estimate the cost of equity.
  - We can often estimate the cost of debt by observing the YTM of the firm's debt.
- ▶ Second, we determine the WACC by weighting these two costs appropriately.

## *Example: WACC*

- Consider a firm whose debt has a market value of \$40 million and whose stock has a market value of \$60 million (3 million outstanding shares of stock, each selling for \$20 per share). The firm pays a 5 percent rate of interest on its new debt and has a beta of 1.41. The corporate tax rate is 34 percent. (Assume that the security market line (SML) holds, that the risk premium on the market is 9.5 percent [somewhat higher than the historical equity risk premium], and that the current Treasury bill rate is 1 percent.)
- **What is this firm's *WACC*?**

## *Example: WACC*

1. The pretax cost of debt is 5 percent, implying an aftertax cost :  
 $= 5\% \times (1 - .34) = 3.3\%$

2. We calculate the cost of equity capital by using the SML:

$$\begin{aligned} R_S &= R_F + \beta (R_M - R_F) \\ &= 1\% + 1.41 \times 9.5\% \\ &= 14.40\% \end{aligned}$$

## *Example: WACC*

3.

$$R_{WACC} = \frac{S}{S+B} \times R_S + \frac{B}{S+B} \times R_B \times (1 - T_C)$$

$$= 60/100 \times 14.40\% + 40/100 \times 3.3\%$$

$$= 9.96\%$$

# Example: Eastman Chemical

**EASTMAN'S COST OF EQUITY** Our first stop for Eastman is [finance.yahoo.com](http://finance.yahoo.com) (ticker symbol: "EMN"). As of March 2015, some of the relevant data are in the next two tables.

Eastman Chemical Co. (EMN) - NYSE ★ Watchlist

**71.41** +2.48 (3.60%) Mar 11, 4:02PM EST

After Hours: **71.41** 0.00 (0.00%) Mar 11, 4:13PM EST

|                |                       |               |               |
|----------------|-----------------------|---------------|---------------|
| Prev Close:    | 68.93                 | Day's Range:  | 69.43 - 71.54 |
| Open:          | 69.71                 | 52wk Range:   | 56.03 - 83.90 |
| Bid:           | 71.27 x 100           | Volume:       | 1,700,064     |
| Ask:           | 71.47 x 200           | Avg Vol (3m): | 1,410,910     |
| 1y Target Est: | 75.80                 | Market Cap:   | 10.55B        |
| Beta:          | 1.44315               | P/E (ttm):    | 12.62         |
| Earnings Date: | Apr 28 - May 2 (Est.) | EPS (ttm):    | 5.66          |
|                |                       | Div & Yield:  | 1.84 (2.62%)  |

To estimate Eastman's cost of equity, we will assume a market risk premium of 7 percent and a risk-free rate of .70%. Eastman's beta on [finance.yahoo.com](http://finance.yahoo.com) is 1.44.

Using Eastman's beta in the CAPM to estimate the cost of equity,<sup>16</sup> we find

$$R_S = .007 + 1.44 \times .07 = .1078, \text{ or } 10.78\%$$

## Trading Information

### Stock Price History

|                                            |        |
|--------------------------------------------|--------|
| Beta:                                      | 1.44   |
| 52-Week Change <sup>3</sup> :              | 3.13%  |
| S&P500 52-Week Change <sup>3</sup> :       | -2.83% |
| 52-Week High (Jun 26, 2015) <sup>3</sup> : | 83.90  |
| 52-Week Low (Jan 29, 2016) <sup>3</sup> :  | 56.03  |
| 50-Day Moving Average <sup>3</sup> :       | 63.69  |
| 200-Day Moving Average <sup>3</sup> :      | 67.64  |

# *Example: Eastman Chemical*

**EASTMAN'S COST OF DEBT** Eastman has 12 long-term bond issues that account for essentially all of its long-term debt. To calculate the cost of debt, we will have to combine these 12 issues and compute a weighted average. We go to <http://finra-markets.morningstar.com/BondCenter/Default.jsp> to find quotes on the bonds. We should note here that finding the yield to maturity for all of a company's outstanding bond issues on a single day is unusual. In our previous discussion on bonds, we found that the bond market is not as liquid as the stock market; on many days, individual bond issues may not trade. To find the book value of the bonds, we go to [www.sec.gov](http://www.sec.gov) and find the most recent 10Q (or 10K) report. The basic information is as follows:

| COUPON RATE | MATURITY | BOOK VALUE (FACE VALUE, IN \$ MILLIONS) | PRICE (% OF PAR) | YIELD TO MATURITY |
|-------------|----------|-----------------------------------------|------------------|-------------------|
| 2.40%       | 2017     | \$999                                   | 100.523%         | 1.96%             |
| 6.30        | 2018     | 167                                     | 104.916          | 2.40              |
| 5.50        | 2019     | 250                                     | 110.208          | 2.61              |
| 2.70        | 2020     | 799                                     | 100.400          | 2.59              |
| 4.50        | 2021     | 250                                     | 104.706          | 3.38              |
| 3.60        | 2022     | 900                                     | 100.938          | 3.43              |
| 7.25        | 2024     | 244                                     | 121.567          | 3.85              |
| 7.625       | 2024     | 54                                      | 124.277          | 3.85              |
| 3.80        | 2025     | 796                                     | 98.858           | 3.95              |
| 7.60        | 2027     | 222                                     | 118.781          | 3.92              |
| 4.80        | 2043     | 497                                     | 91.445           | 5.41              |
| 4.65        | 2044     | 877                                     | 89.162           | 5.40              |

# Example: Eastman Chemical

Comparison of the results of the calculations are as follows:

| COUPON RATE | BOOK VALUE (FACE VALUE, IN \$ MILLIONS) | PERCENTAGE OF TOTAL | MARKET VALUE (IN \$ MILLIONS) | PERCENTAGE OF TOTAL | YIELD TO MATURITY | BOOK VALUES | MARKET VALUES |
|-------------|-----------------------------------------|---------------------|-------------------------------|---------------------|-------------------|-------------|---------------|
| 2.40%       | \$ 999                                  | 16.50               | \$1,004.22                    | 16.52               | 1.96%             | .32%        | .32%          |
| 6.30        | 167                                     | 2.76                | 175.21                        | 2.88                | 2.40              | .07         | .07           |
| 5.50        | 250                                     | 4.13                | 275.52                        | 4.53                | 2.61              | .11         | .12           |
| 2.70        | 799                                     | 13.20               | 802.20                        | 13.20               | 2.59              | .34         | .34           |
| 4.50        | 250                                     | 4.13                | 261.77                        | 4.31                | 3.38              | .14         | .15           |
| 3.60        | 900                                     | 14.86               | 908.44                        | 14.95               | 3.43              | .51         | .51           |
| 7.25        | 244                                     | 4.03                | 296.62                        | 4.88                | 3.85              | .16         | .19           |
| 7.625       | 54                                      | .89                 | 67.11                         | 1.10                | 3.85              | .03         | .04           |
| 3.80        | 796                                     | 13.15               | 786.91                        | 12.95               | 3.95              | .52         | .51           |
| 7.60        | 222                                     | 3.67                | 263.69                        | 4.34                | 3.92              | .14         | .17           |
| 4.80        | 497                                     | 8.21                | 454.48                        | 7.48                | 5.41              | .44         | .40           |
| 4.65        | 877                                     | 14.48               | 781.95                        | 12.86               | 5.40              | .78         | .69           |
| Total       | \$6,055                                 | 100.00              | \$6,078.13                    | 100.00              |                   | 3.57%       | 3.52%         |

# *Example: Eastman Chemical*

**EASTMAN'S WACC** We now have the various pieces necessary to calculate Eastman's WACC. First, we need to calculate the capital structure weights. The market values of Eastman's debt and equity are \$6.078 billion and \$10.55 billion, respectively. The total value of the firm is \$16.628 billion, implying that the debt and equity weights are  $\$6.078/\$16.628 = .366$  and  $\$10.55/\$16.628 = .634$ , respectively. Assuming a tax rate of 35 percent, Eastman's WACC is:

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$$R_{WACC} = .366 \times .0352 \times (1 - .35) + .634 \times .1078 = .0768, \text{ or } 7.68\%$$

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**3. Calculating Cost of Debt** Suspect Corp. issued a 30-year, 5.8 percent semiannual bond seven years ago. The bond currently sells for 104 percent of its face value. The company's tax rate is 35 percent.

- a. What is the pretax cost of debt?
- b. What is the aftertax cost of debt?
- c. Which is more relevant, the pretax or the aftertax cost of debt? Why?

3. a. The pretax cost of debt is the YTM of the company's bonds, so:

$$P_0 = \$1,040 = \$29(PVIFA_{R\%,46}) + \$1,000(PVIF_{R\%,46})$$

$$R = 2.746\%$$

$$R_B = 2 \times 2.746\% = 5.49\%$$

b. The aftertax cost of debt is:

$$\text{Aftertax cost of debt} = 5.49\%(1 - .35)$$

$$\text{Aftertax cost of debt} = 3.57\%$$

c. The aftertax rate is more relevant because that is the actual cost to the company.

**4. Calculating Cost of Debt** For the firm in the previous problem, suppose the book value of the debt issue is \$55 million. In addition, the company has a second debt issue on the market, a zero coupon bond with 12 years left to maturity; the book value of this issue is \$75 million and the bonds sell for 63 percent of par. What is the company's total book value of debt? The total market value? What is your best estimate of the aftertax cost of debt now?

To find the market value of debt, we find the price of the bonds and multiply by the number of bonds. Alternatively, we can multiply the price quote of the bond times the par value of the bonds. Doing so, we find:

$$B = 1.04(\$55,000,000) + .63(\$75,000,000)$$

$$B = \$104,450,000$$

The YTM of the zero coupon bonds is:

$$P_Z = \$630 = \$1,000(\text{PVIF}_{R\%,24})$$

$$R = 1.944\%$$

$$\text{YTM} = 2 \times 1.944\% = 3.89\%$$

So, the aftertax cost of the zero coupon bonds is:

$$\text{Aftertax cost of debt} = 3.89\%(1 - .35)$$

$$\text{Aftertax cost of debt} = 2.53\%$$

$$\text{Aftertax cost of debt} = .0357[1.04(\$55)/\$104.45] + .0253[.63(\$75)/\$104.45]$$

$$\text{Aftertax cost of debt} = .0310, \text{ or } 3.10\%$$

**11. Finding the WACC** Given the following information for Cleen Power Co., find the WACC. Assume the company's tax rate is 35 percent.

*Debt:* 7,000 6 percent coupon bonds outstanding, \$1,000 par value, 20 years to maturity, selling for 105 percent of par; the bonds make semiannual payments.

*Common stock:* 180,000 shares outstanding, selling for \$58 per share; the beta is 1.10.

*Market:* 6.5 percent market risk premium and 4.3 percent risk-free rate.

**11.** We will begin by finding the market value of each type of financing. We find:

$$B = 7,000(\$1,000)(1.05)$$

$$B = \$7,350,000$$

$$S = 180,000(\$58)$$

$$S = \$10,440,000$$

And the total market value of the firm is:

$$V = \$7,350,000 + 10,440,000$$

$$V = \$17,790,000$$

Now, we can find the cost of equity using the CAPM. The cost of equity is:

$$R_S = .043 + 1.10(.065)$$

$$R_S = .1145, \text{ or } 11.45\%$$

The cost of debt is the YTM of the bonds, so:

$$P_0 = \$1,050 = \$30(\text{PVIFA}_{R\%,40}) + \$1,000(\text{PVIF}_{R\%,40})$$

$$R = 2.791\%$$

$$\text{YTM} = 2.791\% \times 2 = 5.58\%$$

And the aftertax cost of debt is:

$$R_B = (1 - .35)(.0558)$$

$$R_B = .0363, \text{ or } 3.63\%$$

Now we have all of the components to calculate the WACC. The WACC is:

$$R_{\text{WACC}} = .0363(\$7,350,000/\$17,790,000) + .1145(\$10,440,000/\$17,790,000)$$

$$R_{\text{WACC}} = .0822, \text{ or } 8.22\%$$

Notice that we didn't include the  $(1 - t_C)$  term in the WACC equation. We used the aftertax cost of debt in the equation, so the term is not needed here.

**13. SML and WACC** An all-equity firm is considering the following projects:

| PROJECT  | BETA | IRR  |
|----------|------|------|
| <i>W</i> | .75  | 9.4% |
| <i>X</i> | .90  | 10.1 |
| <i>Y</i> | 1.15 | 11.9 |
| <i>Z</i> | 1.45 | 14.6 |

The T-bill rate is 3.5 percent, and the expected return on the market is 11 percent.

- Which projects have a higher expected return than the firm's 11 percent cost of capital?
- Which projects should be accepted?
- Which projects would be incorrectly accepted or rejected if the firm's overall cost of capital was used as a hurdle rate?

a. Projects Y and Z.

b. Using the CAPM to consider the projects, we need to calculate the expected return of each project given its level of risk. This expected return should then be compared to the expected return of the project. If the return calculated using the CAPM is lower than the project expected return, we should accept the project; if not, we reject the project. After considering risk via the CAPM:

|          |                            |                             |
|----------|----------------------------|-----------------------------|
| E[W]     | = .035 + .75(.11 – .035)   | = .0913 < .094, so          |
| accept W |                            |                             |
| E[X]     | = .035 + .90(.11 – .035)   | = .1025 > .101, so          |
| reject X |                            |                             |
| E[Y]     | = .035 + 1.15(.11 – .035)  | = .1213 > .119, so reject Y |
| E[Z]     | = .035 + 1.45 (.11 – .035) | = .1438 < .146, so accept Z |

c. Project W would be incorrectly rejected; Project Y would be incorrectly accepted.

**17. Preferred Stock and WACC** The Saunders Investment Bank has the following financing outstanding. What is the WACC for the company?

|                         |                                                                                                                                                                                                                                                                                         |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Debt:</i>            | 90,000 bonds with a coupon rate of 6 percent and a current price quote of 106.4; the bonds have 20 years to maturity. 280,000 zero coupon bonds with a price quote of 15.8 and 30 years until maturity. Both bonds have a par value of \$1,000. Both bonds have semiannual compounding. |
| <i>Preferred stock:</i> | 150,000 shares of 4 percent preferred stock with a current price of \$84, and a par value of \$100.                                                                                                                                                                                     |
| <i>Common stock:</i>    | 2,600,000 shares of common stock; the current price is \$68, and the beta of the stock is 1.15.                                                                                                                                                                                         |
| <i>Market:</i>          | The corporate tax rate is 40 percent, the market risk premium is 7 percent, and the risk-free rate is 3.2 percent.                                                                                                                                                                      |

17. We will begin by finding the market value of each type of financing. We will use B1 to represent the coupon bond, and B2 to represent the zero coupon bond. So, the market value of the firm's financing is:

$$B_{B1} = 90,000(\$1,000)(1.064)$$

$$B_{B1} = \$95,760,000$$

$$B_{B2} = 280,000(\$1,000)(.158)$$

$$B_{B2} = \$44,240,000$$

$$P = 150,000(\$84)$$

$$P = \$12,600,000$$

$$S = 2,600,000(\$68)$$

$$S = \$176,800,000$$

And the total market value of the firm is:

$$V = \$95,760,000 + \$44,240,000 + \$12,600,000 + \$176,800,000$$

$$V = \$329,400,000$$

Now, we can find the cost of equity using the CAPM. The cost of equity is:

$$R_S = .032 + 1.15(.07)$$

$$R_S = .1125, \text{ or } 11.25\%$$

The cost of debt is the YTM of the bonds, so:

$$P_0 = \$1,064 = \$30(\text{PVIFA}_{R\%,40}) + \$1,000(\text{PVIF}_{R\%,40})$$

$$R = 2.735\%$$

$$\text{YTM} = 2.735\% \times 2 = 5.47\%$$

And the aftertax cost of debt is:

$$R_{B1} = (1 - .40)(.0547)$$

$$R_{B1} = .0328, \text{ or } 3.28\%$$

And the aftertax cost of the zero coupon bonds is:

$$P_0 = \$158 = \$1,000(\text{PVIF}_{R\%,60})$$

$$R = 3.123\%$$

$$\text{YTM} = 3.123\% \times 2 = 6.25\%$$

$$R_{B2} = (1 - .40)(.0625) \quad R_{B2} = .0375, \text{ or } 3.75\%$$

$$R_P = D_1/P_0$$

$$R_P = \$4/\$84$$

$$R_P = .0476, \text{ or } 4.76\%$$

Now we have all of the components to calculate the WACC. The WACC is:

$$R_{WACC} = .0328(\$95,760,000/\$329,400,000) + .0375(\$44,240,000/\$329,400,000) \\ + .1125(\$176,800,000/\$329,400,000) + .0476(\$12,600,000/\$329,400,000)$$

$$R_{WACC} = .0768, \text{ or } 7.68\%$$

- 19. Calculating the Cost of Equity** Floyd Industries stock has a beta of 1.25. The company just paid a dividend of \$.95, and the dividends are expected to grow at 4.5 percent per year. The expected return on the market is 11 percent, and Treasury bills are yielding 4.3 percent. The current price of the company's stock is \$64.
- a. Calculate the cost of equity using the DDM method.
  - b. Calculate the cost of equity using the SML method.
  - c. Why do you think your estimates in (a) and (b) are so different?

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19. a. Using the dividend discount model, the cost of equity is:

$$R_S = [(.95)(1.045)/\$64] + .045$$

$$R_S = .0605, \text{ or } 6.05\%$$

b. Using the CAPM, the cost of equity is:

$$R_S = .043 + 1.25(.11 - .043)$$

$$R_S = .1268, \text{ or } 12.68\%$$

c. When using the dividend growth model or the CAPM, you must remember that both are estimates for the cost of equity. Additionally, and perhaps more importantly, each method of estimating the cost of equity depends upon different assumptions.